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Surname:														
First Name(s):														

Department of Mathematics and Applied Mathematics
MAM2000W: 2LA (Linear Algebra)

Test 2

4 May 2022

Time : 1.5 hours

Maximum marks: 52

Full marks: 50

- This question paper consists of 10 pages (including this one).
- Answer all questions in the spaces provided on this question paper. Use the backs of pages when you run out of space.
- Use your *UCT Examination Answer Book* for rough work. We won't take this work in! Only the answers that you write on this question paper will be marked.
- Basic scientific calculators are allowed.
- For Section A, only final answers will be marked. For Section B, full answers must be given.
- Be careful to provide answers that we can read and make sense of at all times. We will pay attention to your presentation as well as the content. Work that is poorly presented will be penalized. If we can't read your handwriting, we will mark your solution incorrect.

DO NOT WRITE BELOW THIS LINE

Q1	Q2	Q3	Q4	Q5	Q6	Q7	Q8	Q9	TOTAL
3	6	3	5	5	6	10	9	5	52

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Section A

In this section, give your final answer only. Do not show your working.

Question 1. [3 points] Let A be a square matrix such that $\det(A) = 5$ and $\det(2A) = 40$. Find

(a) the size of A ;

1pt

3

(b) $\det(A^{-1})$;

1pt

1/5

(c) $\det(A^T A)$.

1pt

$5 \cdot 5 = 25$

Question 2. [6 points] Let $A = \begin{pmatrix} 1 & 3 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & -2 \end{pmatrix}$.

(a) Find all the eigenvalues of A .

3pt

$-2, -2, 4$

(b) Find all the eigenvectors corresponding to the largest eigenvalue of A .

3pt

$s \begin{pmatrix} 1 \\ 1 \\ 0 \end{pmatrix}, s \neq 0$

Question 3. [3 points] Write down a basis for the vector space of all symmetric 3×3 matrices.

3pt

For instance $\left\{ \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix}, \right.$ $\left. \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right\}$
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Question 4. [5 points] Let $A = \begin{pmatrix} 1 & -2 & 1 & 1 & 4 \\ 2 & -4 & 3 & 4 & 9 \\ 3 & -6 & 4 & 5 & 13 \end{pmatrix}$ and $U = \begin{pmatrix} 1 & -2 & 1 & 1 & 4 \\ 0 & 0 & 1 & 2 & 1 \\ 0 & 0 & 0 & 0 & 0 \end{pmatrix}$;

U is a row-echelon form of A (you need not check this).

(a) Find the nullity and rank of A .

2pt

$$\text{Rank}(A) = 2, \text{ Nullity}(A) = 3$$

(b) Find the nullity of A^T .

1pt

$$\text{Nullity}(A^T) = 3 - 2 = 1$$

(c) Write down a basis for (i) $CS(A)$ (ii) $RS(A)$.

(i) A basis for $CS(A)$ is for instance

$$1pt \left\{ \begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 1 \\ 3 \\ 4 \end{pmatrix} \right\}.$$

(ii) A basis for $RS(A)$ is for instance

$$1pt \left\{ (1, -2, 1, 1, 4), (0, 0, 1, 2, 1) \right\}$$

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Section B

In this section, give full answers to all the questions. Show all your working.

Question 5. [5 points] Let V be the vector space of all $n \times n$ matrices and

$$S = \{A \in V : \det(A) = 0\}.$$

Answer the following questions, giving reasons for all your answers.

(a) Is S closed under scalar multiplication?

2pt

Yes, S is closed under scalar multiplication:

Let $A \in S$ and λ be any scalar.

$$\text{Then } \det(\lambda A) = \lambda^n \det(A) = \lambda^n \cdot 0 = 0.$$

It follows that $\lambda A \in S$.

(b) Is S closed under addition?

2pt

S is not closed under addition. There are many counterexamples. Take for instance

$$A = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix} \text{ and } B = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}. \text{ Then } A, B \in S$$

but $\det(A+B) = \det(I_2) = 1$, and hence $A+B \notin S$.

(c) Is S a subspace of V ?

1pt

Because S is not closed under addition, S is not a subspace of V .

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Question 6. [6 points] The function $S : M_{2 \times 2}(\mathbb{R}) \rightarrow M_{2 \times 2}(\mathbb{R})$ is defined by $S(A) = \frac{1}{2}(A + A^T)$.

(a) Show that S is a linear transformation.

2pt

For any $A, B \in M_{2 \times 2}(\mathbb{R})$,

$$\begin{aligned} S(A+B) &= \frac{1}{2}(A+B+(A+B)^T) = \frac{1}{2}(A+B+A^T+B^T) \\ &= \frac{1}{2}(A+A^T) + \frac{1}{2}(B+B^T) = S(A) + S(B) \end{aligned}$$

For any $A \in M_{2 \times 2}(\mathbb{R})$ and scalar λ ,

$$S(\lambda A) = \frac{1}{2}(\lambda A + (\lambda A)^T) = \lambda \cdot \frac{1}{2}(A + A^T) = \lambda \cdot S(A).$$

(b) Find the kernel of S .

2pt

$$\text{We have } A \in \text{Ker}(S) \iff \frac{1}{2}(A + A^T) = 0$$

$$\iff A + A^T = 0$$

$$\iff A^T = -A$$

$$\iff A \text{ is skew symmetric}$$

(c) Find the range of S

2pt

$$S\left(\begin{pmatrix} a & b \\ c & d \end{pmatrix}\right) = \frac{1}{2}\begin{pmatrix} 2a & b+c \\ b+c & 2d \end{pmatrix} = \begin{pmatrix} a & \frac{1}{2}(b+c) \\ \frac{1}{2}(b+c) & d \end{pmatrix}$$

$$\text{Hence } \text{ran}(S) = \left\{ \begin{pmatrix} a & b \\ b & d \end{pmatrix} : a, b, d \in \mathbb{R} \right\}$$

= set of all symmetric matrices

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Question 7. [10 points] Let $S = \{x^3 - x, x^3 + 1, x + 1, x^2\}$.

(a) Is S a linearly independent subset of P_3 ? Give a reason for your answer.

2pt

S is not linearly independent.

It holds that

$$1 \cdot (x^3 - x) + (-1) \cdot (x^3 + 1) + 1 \cdot (x + 1) = 0.$$

(b) Explain why S cannot span P_3 .

1pt

It holds that $\dim(P_3) = 4$.

Hence we need 4 linearly independent vectors to span P_3 .

However, the 4 vectors in S are not linearly independent.

(c) A polynomial $p(x) = ax^3 + bx^2 + cx + d$ will be in $\text{span}(S)$ if and only if the coefficients a, b, c, d satisfy a certain equation. Find this equation.

3pt

The system

$$\begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 \\ -1 & 0 & 1 & 0 \\ 0 & 1 & 1 & 0 \end{pmatrix} \text{ reduces to } \begin{pmatrix} 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{pmatrix} \left| \begin{array}{l} a \\ a+c \\ b \\ d-a-c \end{array} \right.$$

It follows that the equation

$$d - a - c = 0 \text{ must be satisfied.}$$

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- (d) Find a subset B of S that is a basis for $\text{span}(S)$, and explain why it is a basis.

2pt

The vectors corresponding to the pivot columns in the matrix obtained in (c) are $x^3 - x$, $x^3 + 1$, and x^2 . Hence $\{x^3 - x, x^3 + 1, x^2\}$ is a basis.

But any other subset of S of size 3 containing x^2 is also a basis.

- (e) Is it possible to find a polynomial $p(x) \in P_3$ such that $S \cup \{p(x)\}$ is a basis for P_3 ? Give a reason for your answer.

1pt

Not possible because S is linearly dependent.

- (f) Is it possible to find a polynomial $p(x) \in P_3$ such that $B \cup \{p(x)\}$ is a basis for P_3 , where B is the set you found in (d). Give a reason for your answer. If your answer is "yes", give such a $p(x)$.

1pt

It is possible. We can take any polynomial

$$ax^3 + bx^2 + cx + d$$

satisfying $d - a - c \neq 0$.

We can for instance take the polynomial x^3 .

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Question 8. [9 points] Read the following argument and then answer the questions following it.

Let $B = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ be a basis for a vector space V and let $\mathbf{v}$ be a member of V . Then there are scalars $\lambda_1, \lambda_2, \dots, \lambda_n$ such that

$$\mathbf{v} = \lambda_1 \mathbf{u}_1 + \lambda_2 \mathbf{u}_2 + \dots + \lambda_n \mathbf{u}_n. \quad (1)$$

Suppose that we also have $\mathbf{v} = \mu_1 \mathbf{u}_1 + \mu_2 \mathbf{u}_2 + \dots + \mu_n \mathbf{u}_n$ for scalars $\mu_1, \mu_2, \dots, \mu_n$. Then

$$\mathbf{0} = (\lambda_1 - \mu_1) \mathbf{u}_1 + (\lambda_2 - \mu_2) \mathbf{u}_2 + \dots + (\lambda_n - \mu_n) \mathbf{u}_n. \quad (2)$$

Since B is a basis,

$$\lambda_1 - \mu_1 = \lambda_2 - \mu_2 = \dots = \lambda_n - \mu_n = 0. \quad (3)$$

Thus we get that $\lambda_1 = \mu_1, \lambda_2 = \mu_2, \dots, \lambda_n = \mu_n$.

(a) What property of a basis is used to obtain (1)?

A basis for V spans V , i.e. every $\mathbf{v} \in V$ can be written as a linear combination of basis elements.

(1 mark)

(b) How is equation (2) obtained?

$$\begin{aligned} \mathbf{0} &= \mathbf{v} - \mathbf{v} \\ &= \lambda_1 \mathbf{u}_1 + \lambda_2 \mathbf{u}_2 + \dots + \lambda_n \mathbf{u}_n - (\mu_1 \mathbf{u}_1 + \mu_2 \mathbf{u}_2 + \dots + \mu_n \mathbf{u}_n) \\ &= (\lambda_1 - \mu_1) \mathbf{u}_1 + (\lambda_2 - \mu_2) \mathbf{u}_2 + \dots + (\lambda_n - \mu_n) \mathbf{u}_n \end{aligned}$$

(1 mark)

(c) Explain how the fact that B is a basis leads to (3).

Since the basis B is linearly independent, the only linear combination of the basis vectors that can equal the zero vector is the one where all the scalars are equal to 0. Therefore

$\mathbf{0} = (\lambda_1 - \mu_1) \mathbf{u}_1 + (\lambda_2 - \mu_2) \mathbf{u}_2 + \dots + (\lambda_n - \mu_n) \mathbf{u}_n$ implies $\lambda_1 - \mu_1 = \lambda_2 - \mu_2 = \dots = \lambda_n - \mu_n = 0$.

(1 mark)

(d) Carefully state the result proved here.

If B is a basis for a vector space V , every element in V can be written in a unique way as a linear combination of the basis elements.

(1 mark)

(e) Using the notation of the proof, give the coordinate vector $[\mathbf{v}]_B$ of $\mathbf{v}$ with respect to the basis B .

$$[\mathbf{v}]_B = (\lambda_1, \lambda_2, \dots, \lambda_n)$$

(1 mark)

Why is the result proved here needed to define the concept of a coordinate vector?

The result ensures that every vector in V has a unique coordinate vector with respect to the basis B .

(1 mark)

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(f) Prove that for any $\mathbf{v}, \mathbf{w} \in V$, $[\mathbf{v} + \mathbf{w}]_B = [\mathbf{v}]_B + [\mathbf{w}]_B$.

Let $B = \{\mathbf{u}_1, \mathbf{u}_2, \dots, \mathbf{u}_n\}$ be a basis for a vector space V .

Both $\mathbf{v}$ and $\mathbf{w}$ can be written in a unique way as a linear combination of basis elements :

$$\mathbf{v} = \lambda_1 \mathbf{u}_1 + \lambda_2 \mathbf{u}_2 + \dots + \lambda_n \mathbf{u}_n$$

$$\mathbf{w} = \mu_1 \mathbf{u}_1 + \mu_2 \mathbf{u}_2 + \dots + \mu_n \mathbf{u}_n$$

Hence

$$\begin{aligned} \mathbf{v} + \mathbf{w} &= \lambda_1 \mathbf{u}_1 + \lambda_2 \mathbf{u}_2 + \dots + \lambda_n \mathbf{u}_n + \mu_1 \mathbf{u}_1 + \mu_2 \mathbf{u}_2 + \dots + \mu_n \mathbf{u}_n \\ &= (\lambda_1 + \mu_1) \mathbf{u}_1 + (\lambda_2 + \mu_2) \mathbf{u}_2 + \dots + (\lambda_n + \mu_n) \mathbf{u}_n. \end{aligned}$$

Since this must be the unique way in which $\mathbf{v} + \mathbf{w}$ is written as a linear combination of the basis vectors, we get

$$\begin{aligned} [\mathbf{v} + \mathbf{w}]_B &= (\lambda_1 + \mu_1, \lambda_2 + \mu_2, \dots, \lambda_n + \mu_n) \\ &= (\lambda_1, \lambda_2, \dots, \lambda_n) + (\mu_1, \mu_2, \dots, \mu_n) \\ &= [\mathbf{v}]_B + [\mathbf{w}]_B \end{aligned}$$

(3 marks)

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Question 9. [5 points] Let V be a finite dimensional vector space and U be a subspace of V . Show that there is a linear transformation $T : V \rightarrow V$ such that $\ker(T) = U$ and $T(T(\mathbf{v})) = T(\mathbf{v})$ for all $\mathbf{v} \in V$. [Hint: Consider a basis for U . Is there a basis for V containing this basis for U ?]

Let $B_U = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_k\}$ be a basis for U . Because B_U is a linear independent set of vectors in V , B_U can be enlarged to form a basis for V . Let $B_V = \{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_k, \mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_\ell\}$ be a basis for V .

We define the following linear transformation T :

For every $i \in \{1, 2, \dots, k\}$, let $T(\mathbf{b}_i) = \mathbf{0}$, and for every $j \in \{1, 2, \dots, \ell\}$, let $T(\mathbf{c}_j) = \mathbf{c}_j$. (Recall that any linear transformation is uniquely determined by what the linear transformation does to the vectors in the basis.) (3 marks for finding (defining) the correct linear transformation T)

Now we prove that this linear transformation has the required properties.

1. We verify that for every $\mathbf{v} \in V$, it holds that $T(T(\mathbf{v})) = T(\mathbf{v})$.

Let $\mathbf{v}$ be any vector in V . Then there exist scalars $\lambda_1, \lambda_2, \dots, \lambda_k, \mu_1, \mu_2, \dots, \mu_\ell$, such that

$$\mathbf{v} = \lambda_1 \mathbf{b}_1 + \lambda_2 \mathbf{b}_2 + \dots + \lambda_k \mathbf{b}_k + \mu_1 \mathbf{c}_1 + \mu_2 \mathbf{c}_2 + \dots + \mu_\ell \mathbf{c}_\ell.$$

Hence

$$\begin{aligned} T(\mathbf{v}) &= T(\lambda_1 \mathbf{b}_1 + \lambda_2 \mathbf{b}_2 + \dots + \lambda_k \mathbf{b}_k + \mu_1 \mathbf{c}_1 + \mu_2 \mathbf{c}_2 + \dots + \mu_\ell \mathbf{c}_\ell) \\ &= \lambda_1 T(\mathbf{b}_1) + \lambda_2 T(\mathbf{b}_2) + \dots + \lambda_k T(\mathbf{b}_k) + \mu_1 T(\mathbf{c}_1) + \mu_2 T(\mathbf{c}_2) + \dots + \mu_\ell T(\mathbf{c}_\ell) \\ &= \lambda_1 \mathbf{0} + \lambda_2 \mathbf{0} + \dots + \lambda_k \mathbf{0} + \mu_1 \mathbf{c}_1 + \mu_2 \mathbf{c}_2 + \dots + \mu_\ell \mathbf{c}_\ell \\ &= \mu_1 \mathbf{c}_1 + \mu_2 \mathbf{c}_2 + \dots + \mu_\ell \mathbf{c}_\ell. \end{aligned}$$

It follows that

$$\begin{aligned} T(T(\mathbf{v})) &= T(\mu_1 \mathbf{c}_1 + \mu_2 \mathbf{c}_2 + \dots + \mu_\ell \mathbf{c}_\ell) \\ &= \mu_1 T(\mathbf{c}_1) + \mu_2 T(\mathbf{c}_2) + \dots + \mu_\ell T(\mathbf{c}_\ell) \\ &= \mu_1 \mathbf{c}_1 + \mu_2 \mathbf{c}_2 + \dots + \mu_\ell \mathbf{c}_\ell. \end{aligned}$$

Hence we have shown that $T(\mathbf{v}) = T(T(\mathbf{v}))$.

2. We verify that $\ker(T) = U$.

- (a) We show that $U \subseteq \ker(T)$.

Let $\mathbf{u}$ be any vector in U . Since B_U is a basis for U , there exist scalars $\lambda_1, \lambda_2, \dots, \lambda_k$ such that

$$\mathbf{u} = \lambda_1 \mathbf{b}_1 + \lambda_2 \mathbf{b}_2 + \dots + \lambda_k \mathbf{b}_k.$$

It follows that

$$\begin{aligned} T(\mathbf{u}) &= T(\lambda_1 \mathbf{b}_1 + \lambda_2 \mathbf{b}_2 + \dots + \lambda_k \mathbf{b}_k) \\ &= \lambda_1 T(\mathbf{b}_1) + \lambda_2 T(\mathbf{b}_2) + \dots + \lambda_k T(\mathbf{b}_k) \\ &= \lambda_1 \mathbf{0} + \lambda_2 \mathbf{0} + \dots + \lambda_k \mathbf{0} \\ &= \mathbf{0}, \end{aligned}$$

and hence $\mathbf{u} \in \ker(T)$.

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(b) We show that $\ker(T) \subseteq U$.

Let $\mathbf{w}$ be any vector in V .

There exist scalars $\lambda_1, \lambda_2, \dots, \lambda_k, \mu_1, \mu_2, \dots, \mu_\ell$, such that

$$\mathbf{w} = \lambda_1 \mathbf{b}_1 + \lambda_2 \mathbf{b}_2 + \dots + \lambda_k \mathbf{b}_k + \mu_1 \mathbf{c}_1 + \mu_2 \mathbf{c}_2 + \dots + \mu_\ell \mathbf{c}_\ell. \quad (1)$$

Since $\mathbf{w} \in \ker(T)$, it holds that

$$T(\mathbf{w}) = \mathbf{0}.$$

Hence

$$\begin{aligned} T(\mathbf{w}) &= T(\lambda_1 \mathbf{b}_1 + \lambda_2 \mathbf{b}_2 + \dots + \lambda_k \mathbf{b}_k + \mu_1 \mathbf{c}_1 + \mu_2 \mathbf{c}_2 + \dots + \mu_\ell \mathbf{c}_\ell) \\ &= \lambda_1 T(\mathbf{b}_1) + \lambda_2 T(\mathbf{b}_2) + \dots + \lambda_k T(\mathbf{b}_k) + \mu_1 T(\mathbf{c}_1) + \mu_2 T(\mathbf{c}_2) + \dots + \mu_\ell T(\mathbf{c}_\ell) \\ &= \lambda_1 \mathbf{0} + \lambda_2 \mathbf{0} + \dots + \lambda_k \mathbf{0} + \mu_1 \mathbf{c}_1 + \mu_2 \mathbf{c}_2 + \dots + \mu_\ell \mathbf{c}_\ell \\ &= \mu_1 \mathbf{c}_1 + \mu_2 \mathbf{c}_2 + \dots + \mu_\ell \mathbf{c}_\ell \\ &= \mathbf{0} \end{aligned}$$

Since $\mathbf{c}_1, \mathbf{c}_2, \dots, \mathbf{c}_\ell$ are linearly independent, it holds that $\mu_1 = \mu_2 = \dots = \mu_\ell = 0$. Hence Eqn. (1) implies that

$$\mathbf{w} = \lambda_1 \mathbf{b}_1 + \lambda_2 \mathbf{b}_2 + \dots + \lambda_k \mathbf{b}_k.$$

Since $\{\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_k\}$ is a basis for U , it follows that $\mathbf{w} \in U$. □

(2 marks for the correctness proof)